14

Lecture 14

5G Networks: Introduction

مقدمه‌ای بر شبکه‌های نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 5–7 and 9; OFDM basics; Shannon capacity

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-14-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **State** the IMT-2020 usage scenarios and key minimum requirements, and **relate** each to a design choice of 5G.
- **LO2** **Explain** NR's scalable numerology and **compute** slot and symbol durations for each subcarrier spacing.
- **LO3** **Compute** the approximate peak data rate of an NR carrier with the 3GPP formula.
- **LO4** **Compare** non-standalone (EN-DC) and standalone deployments.
- **LO5** **Estimate** latency, connection density and area capacity for given deployments, and **judge** whether a requirement is met.
- **LO6** **Outline** the 5G core and the topics of the following lectures.

2 Prerequisite knowledge

You need the LTE architecture and radio protocols (Lectures 5–7), the EPC (Lecture 9), the structure of an OFDM resource grid (subcarriers, symbols, cyclic prefix, resource blocks) and Shannon's capacity formula $C = B\log_2(1+\mathrm{SNR})$ [11].

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
IMT-2020	IMT-2020	ITU-R's name for the 5G family of radio interfaces and its requirements.
eMBB / URLLC / mMTC	سه سناریوی استفاده	Enhanced broadband, ultra-reliable low latency, massive machine type.
NR	رادیوی جدید	3GPP's 5G radio interface (Release 15 onward).
Numerology μ	عددشناسی	Subcarrier spacing $15 \cdot 2^{\mu}$ kHz and the matching slot length.
FR1 / FR2	بازه‌های بسامدی ۱ و ۲	Sub-7 GHz and millimetre-wave frequency ranges.
Bandwidth part	بخش پهنای باند	Subset of a carrier configured for a UE.
EN-DC	اتصال دوگانهٔ E-UTRA--NR	LTE master and NR secondary node with the EPC (NSA).
SA / NSA	مستقل / غیرمستقل	NR with the 5G core / NR anchored on LTE.
5GC	هستهٔ نسل پنجم	Service-based core network (AMF, SMF, UPF, ...).
Network slice	برش شبکه	Logical end-to-end network for a service type.

3 Introduction and motivation

LTE was designed for one dominant use: mobile broadband to smartphones. By the mid-2010s operators and industries expected three different kinds of demand: far higher capacity for video and fixed wireless access, wireless control of machines with reliability and latency comparable to wired fieldbuses, and connectivity for very large numbers of cheap, battery-powered sensors. ITU-R captured these as the three IMT-2020 usage scenarios (Figure 1) [9] and set minimum technical requirements for any candidate radio interface [10]. 3GPP's answer is the 5G system: the NR radio interface [2], [8] and the service-based 5G core [7].

This lecture opens Part IV. It surveys the requirements, the main ideas of NR that address them, and the migration from LTE; the core network, procedures, slicing and QoS follow in Lectures 18–22, while software-defined networking and virtualisation, the technologies on which the 5G core is built, are the topic of Part III.

4 Fundamental concepts

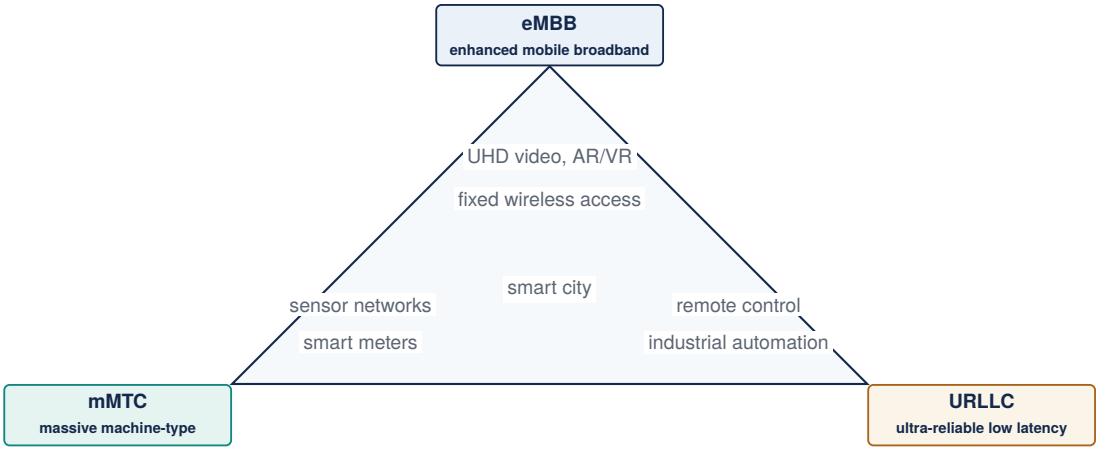


Figure 1. The three IMT-2020 usage scenarios with example applications [9].

Table 1. Selected IMT-2020 minimum technical requirements (ITU-R M.2410) [10].

Requirement	Value	Scenario / remark
Peak data rate	DL 20 Gbit/s, UL 10 Gbit/s	eMBB
Peak spectral efficiency	DL 30, UL 15 bit/s/Hz	eMBB
User-experienced data rate	DL 100 Mbit/s, UL 50 Mbit/s	eMBB, dense urban (5th percentile)
Area traffic capacity	10 Mbit/s/m^2	eMBB, indoor hotspot, DL
User-plane latency	4 ms (eMBB), 1 ms (URLLC)	One way, unloaded
Control-plane latency	20 ms	Idle to active
Connection density	10^6 devices/km^2	mMTC
Reliability	$1 - 10^{-5}$ for 32 bytes in 1 ms	URLLC
Mobility	up to 500 km/h	High-speed trains
Bandwidth	at least 100 MHz; up to 1 GHz above 6 GHz	–

4.1 Requirements

No single deployment has to meet all values at once: each requirement is evaluated in its own test environment. Figure 2 shows the improvement over IMT-Advanced (4G) that ITU-R envisaged for each key capability.

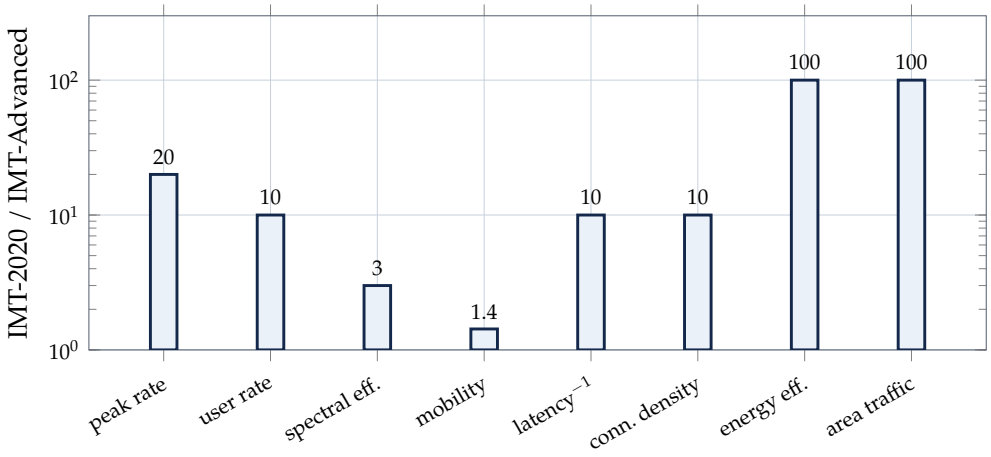


Figure 2. Improvement factors of IMT-2020 key capabilities over IMT-Advanced, as envisaged in ITU-R M.2083 [9]. Latency is shown as its inverse.

4.2 Design principles of NR

- **Wide spectrum range**: FR1 (410 MHz–7.125 GHz) and FR2 (24.25–52.6 GHz, extended in later releases), with carriers up to 100 MHz in FR1 and 400 MHz in FR2 [2].
- **Scalable numerology**: subcarrier spacing $15 \cdot 2^{\mu}$ kHz (Section 6.1).
- **Ultra-lean design**: no always-on cell-specific reference signals; signals are sent only when needed, which saves energy and reduces interference.
- **Beam-centric operation**: synchronisation signal blocks are swept in beams; beam management is built in, which is essential at millimetre waves.
- **Flexible scheduling**: slots and mini-slots (2, 4 or 7 symbols), dynamic TDD, bandwidth parts.
- **New channel coding**: LDPC for data, polar codes for control [3].
- **Forward compatibility**: reserved resources so that future features do not break legacy UEs.

4.3 Deployment options

3GPP defined several ways to combine LTE, NR, the EPC and the 5GC. The two that matter in practice (Figure 3):

- **Non-standalone (NSA), option 3 family (EN-DC)**: the UE is connected to an LTE eNB (master node, control plane to the MME) and an NR en-gNB (secondary node, adding capacity) [1]. It reuses the EPC and allowed the first 5G launches in 2019 without a new core, but provides none of the new core features (slicing, new QoS, service-based architecture).
- **Standalone (SA), option 2**: the gNB connects to the 5GC over N2 (control) and N3 (user plane). All 5G features, including low latency and slicing, require SA.

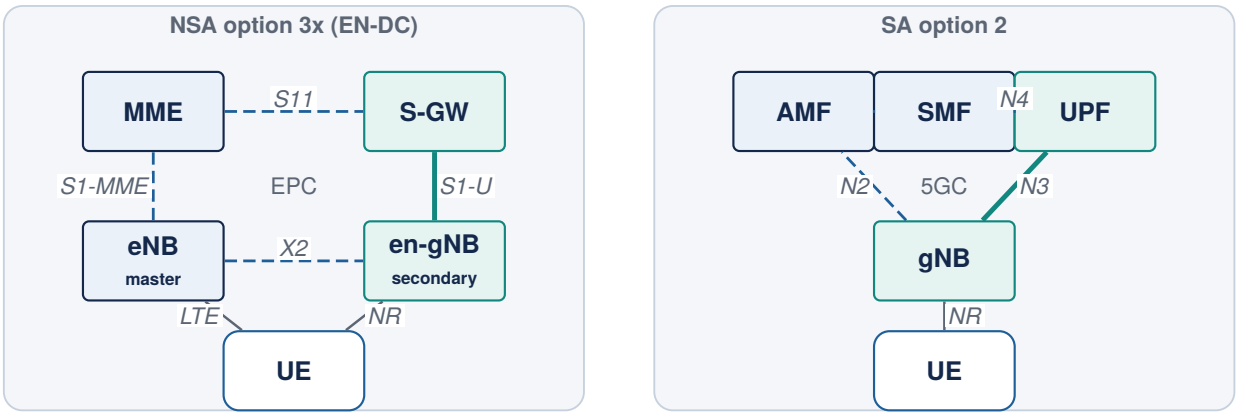


Figure 3. Non-standalone EN-DC (option 3x) with the EPC, and standalone NR (option 2) with the 5GC.

5 Detailed technical explanation

5.1 Frame structure

NR keeps LTE's 10 ms frame and 1 ms subframe. A slot has 14 OFDM symbols (normal cyclic prefix); with numerology μ a subframe contains 2^{μ} slots (Figure 4). A resource block is 12 subcarriers in frequency; a carrier has at most 275 resource blocks. Because 1 ms is a fixed grid for all numerologies, symbols of different numerologies align at subframe boundaries, which allows mixing them on one carrier [3].

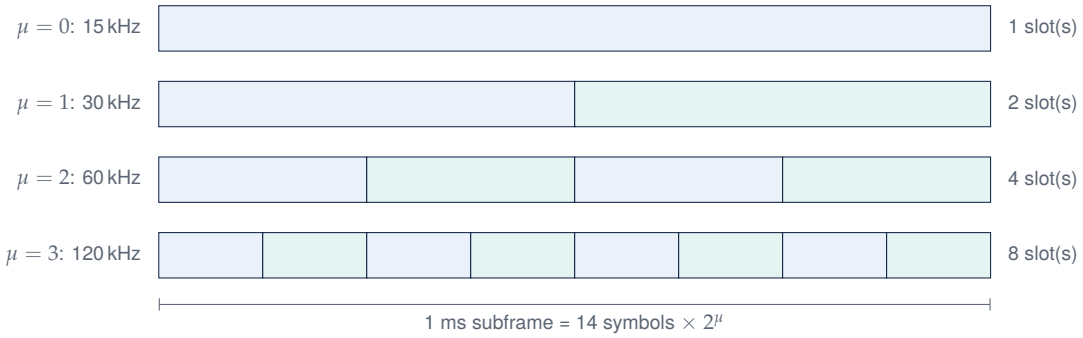


Figure 4. Slots per 1 ms subframe for numerologies $\mu = 0$–3.

5.2 Initial access and beams

A gNB transmits synchronisation signal blocks (SSBs: PSS, SSS and PBCH) in a burst, each in a different beam. The UE detects the best SSB, reads the system information and performs random access on the RACH occasion associated with that SSB; the gNB thereby learns the best beam. During a connection, beams are refined with CSI-RS measurements and reports. In FR1 a few wide beams suffice; in FR2 up to 64 narrow beams compensate the higher path loss with array gain.

5.3 From EPC to 5GC

The 5G core separates access and mobility management (AMF) from session management (SMF) and the user plane (UPF), stores subscription data in the UDM/UDR, authenticates through the AUSF and controls policy in the PCF. Network functions expose services over HTTP/2 and discover each other through the NRF [7]. The QoS model uses QoS flows within PDU sessions instead of EPS bearers, and network slices allow several logical networks on one infrastructure. These topics are developed in Lectures 18–22.

6 Mathematical models

6.1 Numerology

Definitions. For $\mu = 0, 1, 2, 3, 4$:

$$\Delta f = 15 \cdot 2^{\mu} \text{ kHz}, \qquad T_{\text{slot}} = \frac{1 \text{ ms}}{2^{\mu}}, \qquad \bar{T}_{\text{sym}} = \frac{T_{\text{slot}}}{14}, \tag{1}$$

where $\bar{T}_{\text{sym}}$ is the mean symbol duration including the cyclic prefix. The normal cyclic prefix is about $4.7/2^{\mu}$ µs.

Interpretation. Wider spacing shortens symbols and slots (lower latency) and tolerates more phase noise and Doppler (needed in FR2), but its shorter cyclic prefix tolerates less delay spread, which suits small cells. FR1 macro cells typically use 15 or 30 kHz; FR2 uses 120 kHz.

6.2 Peak data rate

Formula. 3GPP approximates the maximum data rate of a carrier as [4]

$$R \approx 10^{-6}\, v\, Q_m\, f\, R_{\max} \frac{N_{\text{PRB}} \cdot 12}{T_s^{\mu}} \left(1 - \text{OH}\right) \text{Mbit/s}, \qquad T_s^{\mu} = \frac{10^{-3}}{14 \cdot 2^{\mu}} \text{ s}, \tag{2}$$

with v MIMO layers, Q_m bits per modulation symbol, scaling factor $f \leq 1$, maximum code rate $R_{\max} = 948/1024$, N_{PRB} resource blocks and overhead OH (0.14 in FR1 DL, 0.18 in FR2 DL, 0.08 in FR1 UL, 0.10 in FR2 UL). For carrier aggregation the rates of the carriers are added.

Example. FR1, 100 MHz at 30 kHz ($N_{\text{PRB}} = 273$), 4 layers, 256-QAM ($Q_m = 8$), $f = 1$: $R \approx 2.34$ Gbit/s (Example 1).

6.3 User-plane latency

Model. A simple estimate of one-way radio latency is

$$T_{\text{UP}} \approx T_{\text{align}} + T_{\text{tx}} + T_{\text{proc}} \approx \tfrac{1}{2} T_{\text{slot}} + T_{\text{slot}} + 2\, T_{\text{slot}}, \tag{3}$$

counting on average half a slot to wait for the next transmission opportunity, one slot of transmission and about two slots of processing at transmitter and receiver.

Example. $\mu = 0$: 3.5 ms; $\mu = 1$: 1.75 ms; $\mu = 3$: 0.44 ms. Mini-slots and pre-configured (grant-free) uplink shorten T_{align} and T_{tx} further.

Limitations. The model ignores scheduling requests, HARQ retransmissions, queueing and the transport network and core; it shows why the URLLC target of 1 ms requires larger numerologies or mini-slots.

6.4 Connection density and area capacity

Models. With device density ρ (devices/km^2) and cell radius r, a cell serves $N = \rho\pi r^2$ devices; if each accesses the network once per period T, the access rate is N/T. For area capacity, a cell with bandwidth B and mean spectral efficiency $\bar{\eta}$ covering area A_{cell} offers

$$C_{\text{area}} = \frac{B\,\bar{\eta}}{A_{\text{cell}}}. \tag{4}$$

Examples. $\rho = 10^6$/km^2, $r = 500$ m: $N = 7.9 \times 10^5$ devices; one report every 2 h gives 109 accesses/s per cell (Example 3). For $C_{\text{area}} = 10$ Mbit/s/m^2 with $B = 400$ MHz and $\bar{\eta} = 5$ bit/s/Hz, each cell may cover only $A_{\text{cell}} = 200$ m^2: area capacity is achieved by densification, not by spectral efficiency alone.

7 Network architecture and diagrams

Figure 3 summarises the two main architectures. In NSA the EPC remains the core, and the NR node is an extension of the LTE radio network (it has no control-plane connection to the core). In SA, the gNB (split into central and distributed units, Lecture 23) connects to the 5GC; LTE and NR then interwork through the core (N26 between AMF and MME) for mobility and voice fallback. Most operators started with NSA and introduced SA later, often first for specific segments such as fixed wireless access or enterprise slices.

8 Step-by-step procedures

Procedure 1 • Adding an NR secondary node in EN-DC

1. The UE is connected to the LTE eNB (master node) and reports NR measurements configured by it.
2. The eNB decides to add an NR leg and sends SgNB Addition Request over X2 to the en-gNB.
3. The en-gNB allocates resources and answers with the NR radio configuration; the eNB forwards it in RRC Connection Reconfiguration.
4. The UE performs random access on the NR cell and confirms; the eNB informs the MME (E-RAB modification) so that the S-GW can send the user plane directly to the en-gNB (option 3x).
5. Data now flow over NR (and LTE, if the bearer is split); mobility and control remain with LTE.

Procedure 2 • Initial access in standalone NR

1. Search the SSB raster, detect PSS/SSS, determine the cell ID and the best SSB beam.
2. Read the MIB on PBCH and SIB1 (scheduled by it) to obtain RACH configuration and cell parameters.
3. Transmit a preamble in the RACH occasion of the best SSB; receive the random-access response (Lecture 8).
4. Complete RRC set-up and send the Registration Request to the AMF (Lecture 20).

9 Worked examples

Worked example 1 • Peak rate of an FR1 carrier

Problem. Evaluate Eq. (2) for 100 MHz at 30 kHz, 4 layers, 256-QAM, $f = 1$, DL.
Solution. $T_s^1 = 10^{-3}/28 = 35.7\,\mu\text{s}$. $N_{\text{PRB}} \cdot 12/T_s = 273 \times 12/35.7 \times 10^{-6} = 9.17 \times 10^7$ REs/s. $R = 4 \times 8 \times 0.926 \times 9.17 \times 10^7 \times 0.86 = 2.34 \times 10^9$ bit/s. To reach 20 Gbit/s, NR needs FR2 bandwidths, more layers and carrier aggregation.

Worked example 2 • Latency and numerology

Problem. Which numerologies satisfy the 1 ms URLLC target with the model of Eq. (3)?
Solution. $T_{\text{UP}} = 3.5\,T_{\text{slot}} \leq 1\,\text{ms}$ requires $T_{\text{slot}} \leq 0.286\,\text{ms}$, i.e. $\mu \geq 2$ (0.875 ms at 60 kHz). With 30 kHz, mini-slots of 2 symbols reduce the transmission and alignment terms and can meet the target as well.

Worked example 3 • mMTC access load

Problem. A cell of radius 500 m in an area with 10^6 devices/km^2; each device sends one report every two hours. Compute the access rate and compare with LTE's typical RACH capacity of the order of a few hundred attempts per second per cell.
Solution. $N = 10^6 \times \pi \times 0.25 = 7.85 \times 10^5$; rate $= 7.85 \times 10^5/7200 = 109\,\text{s}^{-1}$ on average – manageable, but synchronised reporting (all devices at the full hour) would be catastrophic (Lecture 8). mMTC therefore relies on randomised schedules, early data transmission and NB-IoT/LTE-M features, not only on raw capacity.

10 Technology comparison

Evolution, not replacement. LTE continues to evolve (LTE-M, NB-IoT, and dynamic spectrum sharing with NR on the same carrier), and NR reuses many LTE concepts: OFDM, HARQ, the layer-2 stack, and the attach/registration and bearer/flow logic of the core.

11 Key takeaways

Table 2. LTE and NR compared.

	LTE (4G)	NR (5G)
Spectrum	Below 6 GHz	FR1 and FR2 (mmWave)
Max carrier bandwidth	20 MHz (aggregation up to 32 carriers)	100 MHz (FR1), 400 MHz (FR2)
Subcarrier spacing	15 kHz (fixed)	$15 \cdot 2^{\mu}$ kHz
Scheduling unit	1 ms subframe	Slot ($1/2^{\mu}$ ms), mini-slot
Reference signals	Always-on CRS	On demand (ultra-lean)
Channel coding	Turbo (data), convolutional (control)	LDPC (data), polar (control)
Beamforming	Added in later releases	Native, including initial access
Core	EPC	EPC (NSA) or 5GC (SA)
QoS unit	EPS bearer	QoS flow

Key takeaways

- IMT-2020 defines three scenarios (eMBB, URLLC, mMTC) and minimum requirements such as 20 Gbit/s peak, 1 ms URLLC latency and 10^6 devices/km^2.
- NR: wide spectrum (FR1, FR2), numerology $15 \cdot 2^{\mu}$ kHz, slots of $1/2^{\mu}$ ms, lean and beam-based.
- Peak rate $\propto v\, Q_m\, R_{\max}\, N_{\mathrm{PRB}} \cdot 12/T_s\,(1-\mathrm{OH})$; 100 MHz FR1 gives about 2.3 Gbit/s with 4 layers.
- NSA (EN-DC) reuses the EPC; SA with the 5GC unlocks slicing, new QoS and low latency.
- Area capacity comes from densification; latency from numerology and mini-slots; density from access design.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which scenario?

Assign each service to eMBB, URLLC or mMTC and name its most demanding requirement: (a) 8K video streaming; (b) motion control in a factory; (c) water meters; (d) remote surgery; (e) fixed wireless access to homes.

Exercise 2 • Level 1 – conceptual • Why NSA first?

Give two reasons why most operators launched 5G as NSA, and one capability they could not offer until SA.

Exercise 3 • Level 2 – numerical • FR2 peak rate

Compute the peak DL rate of a 400 MHz FR2 carrier at 120 kHz ($N_{\mathrm{PRB}} = 264$) with 2 layers and 64-QAM. How many such carriers would be needed for 20 Gbit/s?

Exercise 4 • Level 2 – numerical • Cyclic prefix and cell size

The normal cyclic prefix is about $4.7/2^{\mu}$ µs. Compute it for $\mu = 0$–3 and the corresponding path-length difference it can absorb. Which numerologies suit a rural macro cell with a delay spread of 2 µs?

Exercise 5 • Level 3 – analytical • Peak spectral efficiency

Using Eq. (2), compute the peak spectral efficiency of a 100 MHz FR1 carrier at 30 kHz with 8 DL layers and 256-QAM. Does it meet the IMT-2020 requirement of 30 bit/s/Hz? Which assumptions make this a theoretical value?

Exercise 6 • Level 3 – analytical • Densification

An indoor hotspot must provide 10 Mbit/s/m^2. Compare the cell area needed with (a) 400 MHz at 5 bit/s/Hz and (b) 100 MHz at 7 bit/s/Hz. Discuss the implications for backhaul and for the core network.

Solutions

Solution to Exercise 1

(a) eMBB – user data rate; (b) URLLC – latency and reliability; (c) mMTC – connection density and battery life; (d) URLLC (with eMBB video) – reliability and latency; (e) eMBB – capacity per area and per user.

Solution to Exercise 2

NSA reused the existing EPC (no new core, faster time to market) and LTE coverage for control and mobility, while NR added capacity; early devices and spectrum were also tuned for it. Features needing the 5GC – network slicing, QoS flows, service-based exposure, low-latency edge user planes, and NR-only coverage – required SA.

Solution to Exercise 3

$T_s^3 = 10^{-3}/112 = 8.93$ µs; $R = 2 \times 6 \times 0.926 \times 264 \times 12/8.93 \times 10^{-6} \times 0.82 = 3.23$ Gbit/s. For 20 Gbit/s about $20/3.23 = 6.2$, i.e. 7 carriers – or more layers and higher-order modulation.

Solution to Exercise 4

$\mu = 0$: 4.69 µs (1406 m); $\mu = 1$: 2.34 µs (703 m); $\mu = 2$: 1.17 µs (352 m); $\mu = 3$: 0.59 µs (176 m), using $d = c\,T_{\mathrm{CP}}$. With 2 µs delay spread, $\mu = 0$ (and marginally $\mu = 1$) are suitable; 60 kHz needs the extended CP, and 120 kHz is for small FR2 cells.

Solution to Exercise

With 8 layers the rate doubles to 4.67 Gbit/s, i.e. 46.7 bit/s/Hz over 100 MHz – above 30. The value assumes 8 spatially separable layers, 256-QAM at the highest code rate (very high SINR), the fixed overhead factor and full allocation to one UE; real systems reach a fraction of it.

Solution to Exercise

(a) $400 \times 10^6 \times 5 = 2\,\text{Gbit/s}$ per cell, so $A = 2 \times 10^9 / 10^7 = 200\,\text{m}^2$; (b) 0.7 Gbit/s, so $70\,\text{m}^2$. Either way cells are rooms, not streets. Each cell needs multi-gigabit backhaul (fibre), and the core must handle traffic of 10 Tbit/s per km^2 of such hotspots locally – an argument for edge user planes (Lecture 23).

13 Further reading

ITU-R M.2083 [9] sets out the vision and M.2410 [10] the requirements; 3GPP's own scenarios are in TR 38.913 [6] and service requirements in TS 22.261 [5]. NR is described in TS 38.300 [2] and TS 38.211 [3], EN-DC in TS 37.340 [1] and the 5G system in TS 23.501 [7]. Dahlman et al. [8] is the standard textbook on NR.

References

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